THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 4$, $DB = 3$, $AE = 8$. Then $EC =$ 1

(a) 5 **(b)** 7 **(c)** 8 **(d)** 6

2. The areas of two similar triangles are in the ratio 16:25. The ratio of their corresponding sides is 4: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

3. The diagonal of a rectangle with sides 7 and 24 is: 1

(a) 27 **(b)** 25 **(c)** 24 **(d)** 26

4. A triangle has sides 8, 15, 17. It is right-angled if: 1

(a) $8 + 15 = 17$ **(b)** $8^2 + 15^2 = 17^2$ **(c)** $8 \times 15 = 17$ **(d)** $8^2 = 15^2 + 17^2$

5. In a right triangle with legs 20 and 21, the hypotenuse is: 1

(a) 31 **(b)** 29 **(c)** 30 **(d)** 28

6. The ratio of corresponding sides of two similar triangles is 5:6. Their perimeters are in the ratio 5: 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 8

7. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$, $DB = 7$, $AE = 4$. Then $EC =$ 1

(a) 15 **(b)** 14 **(c)** 16 **(d)** 13

8. The areas of two similar triangles are in the ratio 4:49. The ratio of their corresponding sides is 2: 1

(a) 9 **(b)** 7 **(c)** 6 **(d)** 8

9. The diagonal of a rectangle with sides 20 and 21 is: 1

(a) 28 **(b)** 31 **(c)** 29 **(d)** 30

10. A triangle has sides 7, 24, 25. It is right-angled if: 1

(a) $7 \times 24 = 25$ **(b)** $7 + 24 = 25$ **(c)** $7^2 = 24^2 + 25^2$ **(d)** $7^2 + 24^2 = 25^2$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $DE \parallel BC$, $AD = 4$ cm, $DB = 3$ cm, $AE = 8$ cm. Find EC. 2

12. Find the hypotenuse of a right triangle with legs 15 cm and 20 cm. 2

13. The sides of two similar triangles are in the ratio 4:7. Find the ratio of their areas. 2

14. In $\triangle ABC$, $DE \parallel BC$, $AD = 5$ cm, $DB = 6$ cm, $AE = 10$ cm. Find EC. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a right triangle the two perpendicular sides are 20 cm and 21 cm. Find the hypotenuse and the perimeter. 3

16. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 5$ cm, $DB = 2$ cm and $AE = 10$ cm, find EC. 3

17. The sides of a triangle are 4 cm, 6 cm and 8 cm. Is it a right triangle? If so, name the right angle. 3

OR

In a right triangle the two perpendicular sides are 9 cm and 40 cm. Find the hypotenuse and the perimeter. 3

18. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 3$ cm, $DB = 7$ cm and $AE = 9$ cm, find EC. 3

OR

The sides of a triangle are 12 cm, 35 cm and 37 cm. Is it a right triangle? If so, name the right angle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ladder 15 m long reaches a window 12 m above the ground. Find the distance of the foot of the ladder from the wall. 5

OR

A pole 7 m high casts a shadow 2 m long. At the same time a tower casts a shadow 18 m long. Find the height of the tower. 5

20. A ladder 20 m long reaches a window 16 m above the ground. Find the distance of the foot of the ladder from the wall. 5

OR

A pole 3 m high casts a shadow 2 m long. At the same time a tower casts a shadow 20 m long. Find the height of the tower. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND